

益

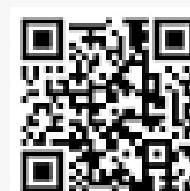
(1) $5x^3 - 2x + 9$ 多项式

各项: $5x^3$, $-2x$, 9

各项系数: 5 , -2 , 9

常数项: 9

次数: 3



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$$(2) \quad a = -1, b = 3$$

$$2a^2 - ab + b$$

求值

$$= 2 \times (-1)^2 - (-1) \times 3 + 3$$

→ 数字

$$= 2 \times 1 + 3 + 3$$
$$= 8$$



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$$\begin{aligned}
 & (3) \quad \underline{7x^2 - 3x + 5} - \underline{2x^2 + 8x - 9} \quad a = -1, b = 3 \\
 & \quad \quad \quad 2a^2 - ab + b \\
 & = (7x^2 - 2x^2) + (-3x + 8x) + (5 - 9) = 2 \times (-1)^2 - (-1) \times 3 + 3 \\
 & = 5x^2 + 5x - 4 \quad = 2 \times 1 + 3 + 3 \\
 & \quad \quad \quad = 8
 \end{aligned}$$



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$$\begin{aligned}
 (3) \quad & \underline{7x^2 - 3x + 5} - \underline{2x^2 + 8x - 9} \\
 &= (7x^2 - 2x^2) + (-3x + 8x) + (5 - 9) \\
 &= 5x^2 + 5x - 4
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad & 4(2x - 3) - 2(3x + 1) \\
 &= 4 \times 2x - 4 \times 3 \\
 &\quad - 2 \times 3x - 2 \times 1 \\
 &= \underline{8x} - 12 - \underline{6x} - \underline{2} \\
 &= \underline{2x} - 14
 \end{aligned}$$



$$\begin{aligned}
 (5) \quad & 3a - \{2a - [a - 2(a-b)]\} \\
 &= 3a - \{2a - [a - (2a - 2b)]\} \\
 &= 3a - \{2a - [a - 2a + 2b]\} \\
 &= 3a - \{2a - [-a + 2b]\} \\
 &= 3a - \{2a + a - 2b\} \\
 &= 3a - \{3a - 2b\} \\
 &= 3a - 3a + 2b = 2b
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad & 4(2x-3) - 2(3x+1) \\
 &= 4 \times 2x - 4 \times 3 \\
 &\quad - 2 \times 3x - 2 \times 1 \\
 &= 8x - 12 - 6x - 2 \\
 &= 2x - 14
 \end{aligned}$$



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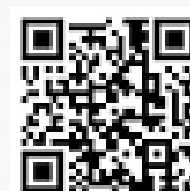
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$$\begin{aligned}
 (5) \quad & 3a - \{2a - [a - 2(a-b)]\} \\
 &= 3a - \{2a - [a - (2a - 2b)]\} \\
 &= 3a - \{2a - [a - 2a + 2b]\} \\
 &= 3a - \{2a - [-a + 2b]\} \\
 &= 3a - \{2a + a - 2b\} \\
 &= 3a - \{3a - 2b\} \\
 &= 3a - 3a + 2b = 2b
 \end{aligned}$$

$$(6) \quad 2x^2y \text{ 和 } -5xy^2 \text{ 同类型项?}$$

不是。

$$\begin{aligned}
 &\checkmark 2x^2y \text{ 与 } -10x^2y \\
 &\checkmark 5xy^2 \text{ 与 } 100xy^2 \\
 &\checkmark 5xy - 10xy = -10yx
 \end{aligned}$$



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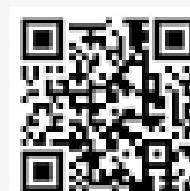
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(2) |8| 绝对值

$$|+4.3| = 4.3$$

$$|-3| = 3$$

$$\begin{aligned} & |-6| + |2-7| - |-3| \quad \begin{array}{l} \text{负数} \\ < 0 \end{array} \\ &= 6 + |-5| - 3 \\ &= 6 + 5 - 3 = 8 \end{aligned}$$



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指数

$$(10) \quad (3^4) \times 3^{-2}$$

$$= (3 \times 3 \times 3 \times 3) \times \frac{1}{3 \times 3}$$

$$= 81 \times \frac{1}{9}$$

$$= 9$$

$$\frac{x^8}{x^3} = \frac{\cancel{x} \cdot \cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x \cdot x \cdot x}{\cancel{x} \cdot \cancel{x} \cdot \cancel{x}}$$

$$= x^5$$

$$\frac{x^{\square}}{x^{\triangle}} = x^{\square - \triangle}$$



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$$(10) (2a^2b)^3$$

$$= (2a^2b) \times (2a^2b) \times (2a^2b)$$

$$= (2 \times 2 \times 2) \times (a^2 a^2 a^2) \times b b b$$

$$= 8a^6b^3$$

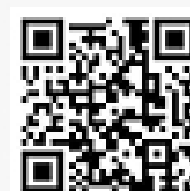
$$(2a^2b)^3 = 2^3 (a^2)^3 b^3$$

$$= 8a^6b^3$$

$$\frac{x^8}{x^3} = \frac{\cancel{x} \cdot \cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x \cdot x \cdot x}{\cancel{x} \cdot \cancel{x} \cdot \cancel{x}}$$

$$= x^5$$

$$\frac{x^{\square}}{x^{\triangle}} = x^{\square - \triangle}$$



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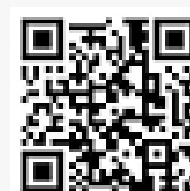
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$$\begin{aligned}
 (10) \quad (2a^2b)^3 &= (2a^2b) \times (2a^2b) \times (2a^2b) \\
 &= (2 \times 2 \times 2) \times (a^2 a^2 a^2) \times b b b \\
 &= 8a^6b^3
 \end{aligned}$$

$$\begin{aligned}
 (2a^2b)^3 &= 2^3 (a^2)^3 b^3 \\
 &= 8a^6b^3
 \end{aligned}$$

$$\begin{aligned}
 (11) \quad \sqrt{32} &= \sqrt{16 \times 2} \\
 &= \sqrt{16} \times \sqrt{2} \\
 &= 4\sqrt{2}
 \end{aligned}$$

根号 $3 \times 3 = 9$
 $\sqrt{9} = 3$
 $100 \times 100 = 10000$
 $\sqrt{10000} = 100$
 $4 \times 4 = 16$



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$$\sqrt{9} = -3?$$

$$\sqrt{(-3) \times (-3)} = \sqrt{9} = 3 = |-3|$$

$$3 \times 3 = 9$$

$$\sqrt{9} = 3$$

$$\sqrt{9} \neq -3$$

✓ 正数
 > 0

负数
 < 0

$$(11) \sqrt{12a^2}$$

$$= \sqrt{12} \times \sqrt{a^2}$$

$$= \sqrt{4 \times 3} \times \sqrt{a^2}$$

$$= 2\sqrt{3}|a| \text{ 不是 } a$$

因为

$$a < 0$$

要取 $|a|$



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(14) 合并同类项

$$\begin{aligned} & \underline{\underline{4x^2 - 3x + 7}} - \underline{\underline{9x^2 + 5x + 2}} \\ &= \underline{\underline{(4x^2 - 9x^2)}} + \underline{\underline{(-3x + 5x)}} + \underline{\underline{9}} \\ &= -5x^2 + 2x + 9 \end{aligned}$$

(13) $x = -2, y = 3$

$$\begin{aligned} & 2x^2 - xy + y \\ &= 2 \times (-2)^2 - (-2) \times 3 + 3 \\ &= 2 \times 4 + 2 \times 3 + 3 \\ &= 8 + 6 + 3 = 17 \end{aligned}$$



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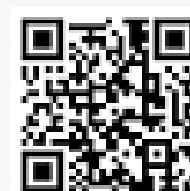
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(14) 合并同类项

$$\begin{aligned} & \underline{4x^2 - 3x + 7} - \underline{9x^2 + 5x - 2} \\ &= \underline{(4x^2 - 9x^2)} + \underline{(-3x + 5x)} + \underline{5} \\ &= -5x^2 + 2x + 5 \end{aligned}$$

(13) $x = -2, y = 3$

$$\begin{aligned} & 2x^2 - xy + y \\ &= 2 \times (-2)^2 - (-2) \times 3 + 3 \\ &= 2 \times 4 + 2 \times 3 + 3 \\ &= 8 + 6 + 3 = 17 \end{aligned}$$



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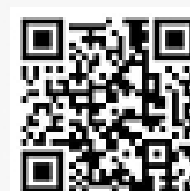
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(14) 合并同类项

$$\begin{aligned} &= \underline{4x^2 - 3x + 7} - \underline{9x^2 + 5x - 2} \\ &= \underline{(4x^2 - 9x^2)} + \underline{(-3x + 5x)} + \underline{5} \\ &= -5x^2 + 2x + 5 \end{aligned}$$

(15)

$$\begin{aligned} &3(2x - 5) - 2(x + 4) \\ &= 6x - \underbrace{3 \times 5}_{15} \\ &\quad - 2x - \underbrace{2 \times 4}_{8} \\ &= 4x - 23 \end{aligned}$$

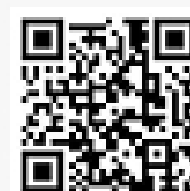


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$$\begin{aligned}
 (16) \quad & (x+5)(x-2) \\
 &= x(x-2) + 5(x-2) \\
 &= x^2 - 2x + 5x - 5 \times 2 \\
 &= x^2 + 3x - 10
 \end{aligned}$$

$$\begin{aligned}
 (17) \quad & (2x-3)^2 \\
 &= (2x-3)(2x-3) \\
 &= 2x(2x-3) - 3(2x-3) \\
 &= (2x)(2x) - 2x \times 3 - 3 \times 2x + 9 \\
 &= 4x^2 - 6x - 6x + 9 \\
 &= 4x^2 - 12x + 9
 \end{aligned}$$



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B组

$$(1) a = -\frac{1}{2}, b = 3$$

$$\frac{2a^2 - ab}{a - b}$$

$$= \frac{2 \times \left(-\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right) \times 3}{-\frac{1}{2} - 3}$$

$$= \frac{\frac{1}{2} + \frac{3}{2}}{-\frac{7}{2}}$$

$$= \frac{2}{-\frac{7}{2}} = 2 \times \left(-\frac{2}{7}\right) = -\frac{4}{7}$$

$ab = 1 \Rightarrow a, b$ 倒数

$$(17) (2x - 3)^2$$

$$= (2x - 3)(2x - 3)$$

$$= 2x(2x - 3) - 3(2x - 3)$$

$$= (2x)(2x) - 2x \times 3 - 3 \times 2x + 9$$

$$= 4x^2 - 6x - 6x + 9$$

$$= 4x^2 - 12x + 9$$



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B组

$$-\sqrt{5} < -2.1 \quad \begin{array}{r} 2.1 \\ \times 2.1 \\ \hline \end{array}$$

$$\sqrt{5} > 2.1 \quad \begin{array}{r} 2.1 \\ \times 2.1 \\ \hline \end{array}$$

$$\sqrt{5} \times \sqrt{5} > 2.1 \times 2.1 \quad \begin{array}{r} 2.1 \\ \times 2.1 \\ \hline 4.41 \end{array}$$

$$5 > 4.41$$

$$-\frac{9}{4} < -\sqrt{5} < -2.1 < 0 < \sqrt{4}$$

负数 < 0

正数 > 0

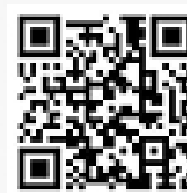
$$(2) \quad -\sqrt{5}, -2.1, -\frac{9}{4}, 0, \sqrt{4}=2$$

$$-\sqrt{5} > -2.25 \quad -2.25 \text{ 最小}$$

$$\sqrt{5} < 2.25 \quad \begin{array}{r} 2.25 \\ \times 2.25 \\ \hline \end{array}$$

$$\sqrt{5} \times \sqrt{5} < 2.25 \times 2.25 \quad \begin{array}{r} 2.25 \\ \times 2.25 \\ \hline 5.0625 \end{array}$$

$$5 < 5.0625$$



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B组

$$-\sqrt{5} < -2.1$$

$$\sqrt{5} > 2.1$$

$$\sqrt{5} \times \sqrt{5} > 2.1 \times 2.1$$

$$5 > 4.41$$

$$-\frac{9}{4} < -\sqrt{5} < -2.1 < 0 < \sqrt{4} = \frac{4}{9} + 1 = \frac{13}{9}$$

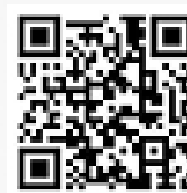
$$(3) \left(\frac{27}{8}\right)^{-\frac{2}{3}} + 16^{\frac{3}{4}} - \sqrt{49}$$

$$= \left(\sqrt[3]{\frac{27}{8}}\right)^{-2} + \left(\sqrt[4]{16}\right)^3 - 7$$

$$= \left(\frac{3}{2}\right)^{-2} + 2^3 - 7$$

$$= \left(\frac{2}{3}\right)^2 + 8 - 7$$

$$= \frac{4}{9} + 1 = \frac{13}{9}$$



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B组

$$(7) (2x-1)(x+3) - (x-2)^2$$

$$= 2x^2 + 6x - x - 3 - (x-2)^2$$

$$= 2x^2 + 5x - 3 - (x-2)(x-2)$$

$$= 2x^2 + 5x - 3 - (x^2 - 4x + 4)$$

$$= 2x^2 + 5x - 3 - x^2 + 4x - 4$$

$$= x^2 + 9x - 7$$

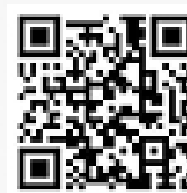
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$$= \frac{4}{9} + 1 = \frac{13}{9}$$



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